CivicPulse

Team Contributions from Steps 1-7

Amna - Steps 1-3 and Submission

- Defined the CivicPulse problem and project objectives.
- Helped establish the initial data sources and data pipeline.
- Contributed to the Bronze, Silver, and Gold data layers.
- Supported data cleaning, validation, provenance, and documentation.
- Completed and submitted the final project materials.

Asma - Steps 3-4 and Demo Video

- Added and integrated Open-Meteo weather features.
- Implemented weather validation, manifest tracking, and warehouse integration.
- Connected weather data to the Gold hourly timeline.
- Built and evaluated the weekly baseline and explainable Poisson model.
- Added model testing, prediction intervals, grouped error analysis, and hourly retraining.
- Created the project demonstration video.

Asiya - Steps 5-7 and Presentation

- Developed the retrieval/RAG layer using the validated Gold data and source provenance.
- Designed the system to retrieve relevant civic reports, issue categories, locations, timestamps, and supporting source links.
- Built the generative layer to convert retrieved information into clear civic summaries and responses.
- Ensured generated answers remained grounded in retrieved evidence and clearly communicated uncertainty.
- Created the user-facing application or dashboard.
- Supported deployment, monitoring, scheduled refreshes, and workflow reliability.
- Created the final PowerPoint presentation.